

AC9M6N08 • Year 6 Mathematics

Estimating with Rational Numbers and Percentages

Choose bounds, compatible numbers and safe financial estimates

We are learning to...

Students select rounding, benchmark fractions, compatible numbers and upper or lower bounds according to the decision, then explain expected estimation error.

Represent

Reason

Calculate

Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

approximate numerical solutions to problems involving rational numbers and percentages, including financial contexts, using appropriate estimation strategies

Estimate a percentage and decimal calculation

calculation	purposeful estimate
18.7% of 398	about 20% of 400 = 80
63.4 ÷ 7.9	about 64 ÷ 8 = 8
\$29.95 + \$18.60 + \$7.85	safe budget: \$30 + \$19 + \$8 = \$57
3/8 of 79	about 3/8 of 80 = 30

The estimate should be simple enough to calculate and close enough for the decision. State whether it is likely high or low.

Use upper and lower bounds for decisions

cash needed

round costs upward

minimum capacity

round demand upward

quick reasonableness

round to one useful place

range estimate

calculate plausible lower and upper values

A balanced estimate is not always safest. Financial sufficiency and capacity decisions often require a deliberate upper bound.

Build and annotate the model

Represent estimating with rational numbers and percentages and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Nearest rounding used automatically:** Choose strategy from purpose.
- **Estimate reported as exact:** Use approximation language and precision.
- **Bounds direction ignored:** A safe budget estimate should not underestimate.
- **Early rounding compounds error:** Keep exact values when later accuracy matters.

Quick check

- Estimate 18.7% of 398.
- Use compatible numbers for division.
- Make a safe budget.
- State high/low expectation.
- Give an estimation range.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.